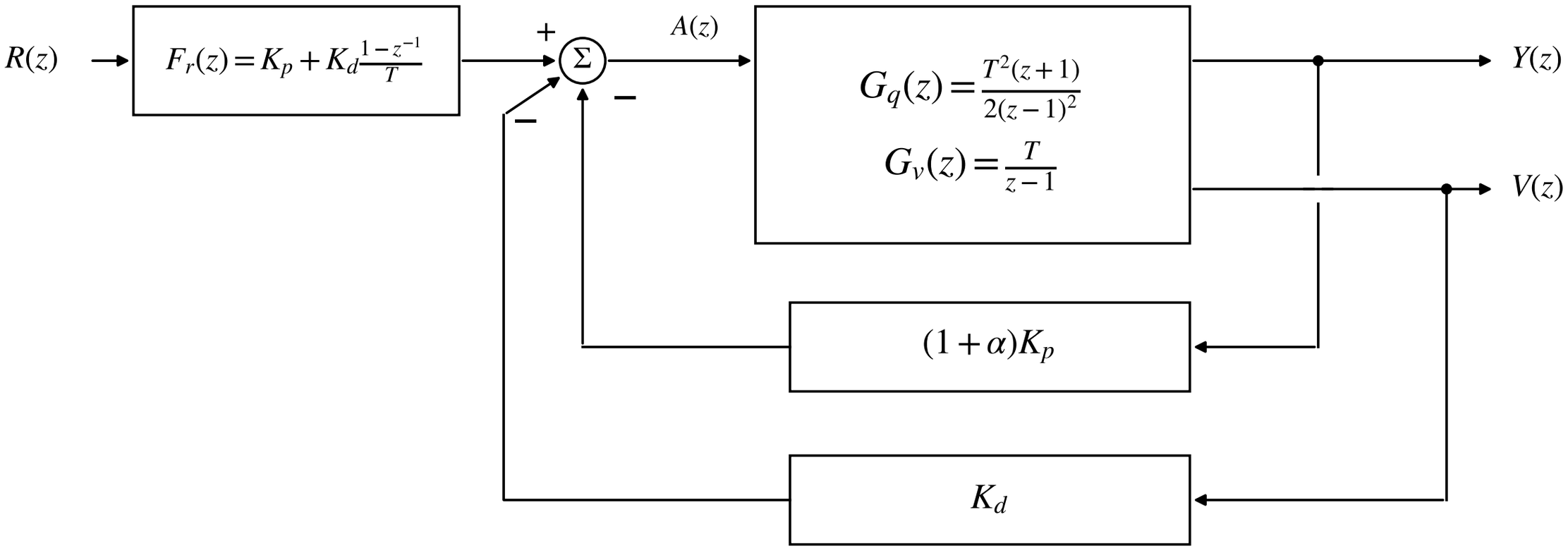
V16 | z-domain feedback: ideal acceleration-ZOH channel

T = 0.020 s. Ideal acceleration hold after compensation; actual V16 holds torque. Measured velocity is a separate feedback.



$$\frac{Y(z)}{R(z)} = \frac{G_q(z)F_r(z)}{1 + (1+\alpha)K_p G_q(z) + K_d G_v(z)}$$

$$D_{\mathrm{cl}}(z) = (z-1)^2 + K_d T(z-1) + \tfrac{1}{2}(1+\alpha)K_p T^2(z+1)$$

Contact QP / 571: Kp = 319.305 s^-2, Kd = 27.468 s^-1, alpha = 0.529243
Whole-body QP / 577: Kp = 289.332 s^-2, Kd = 27.173 s^-1, alpha = 0.647501

Dcl is the homogeneous feedback characteristic; Fr adds reference memory z^-1. Reference velocity / acceleration clipping and QP changes are inactive.